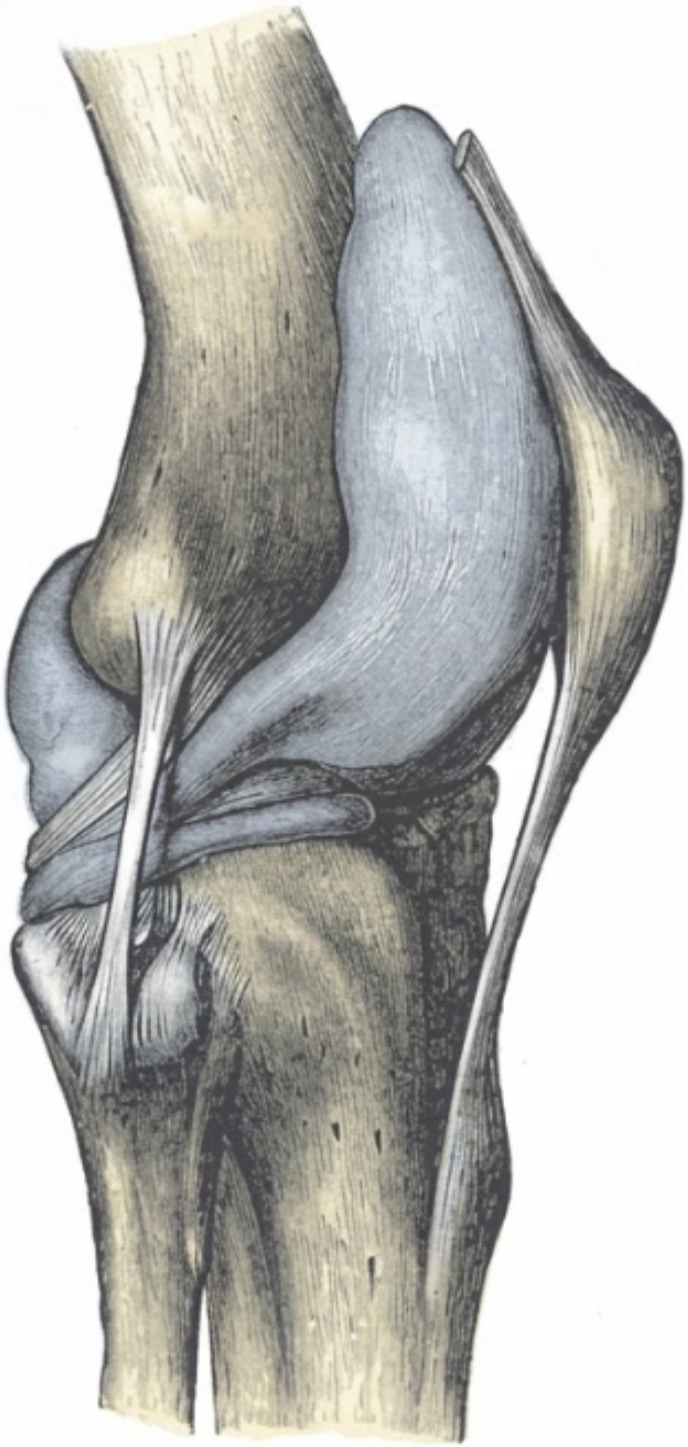
Knee Joint



Learning Objectives

Know the muscles which extend and flex the knee

Understand how the knees locks when fully extended

Appreciate the role of popliteus in unlocking the knee

Enumerate the 4 important ligaments of the knee

Describe the knee capsule, bursae and 2 menisci of the knee

Interpret X-rays of the knee

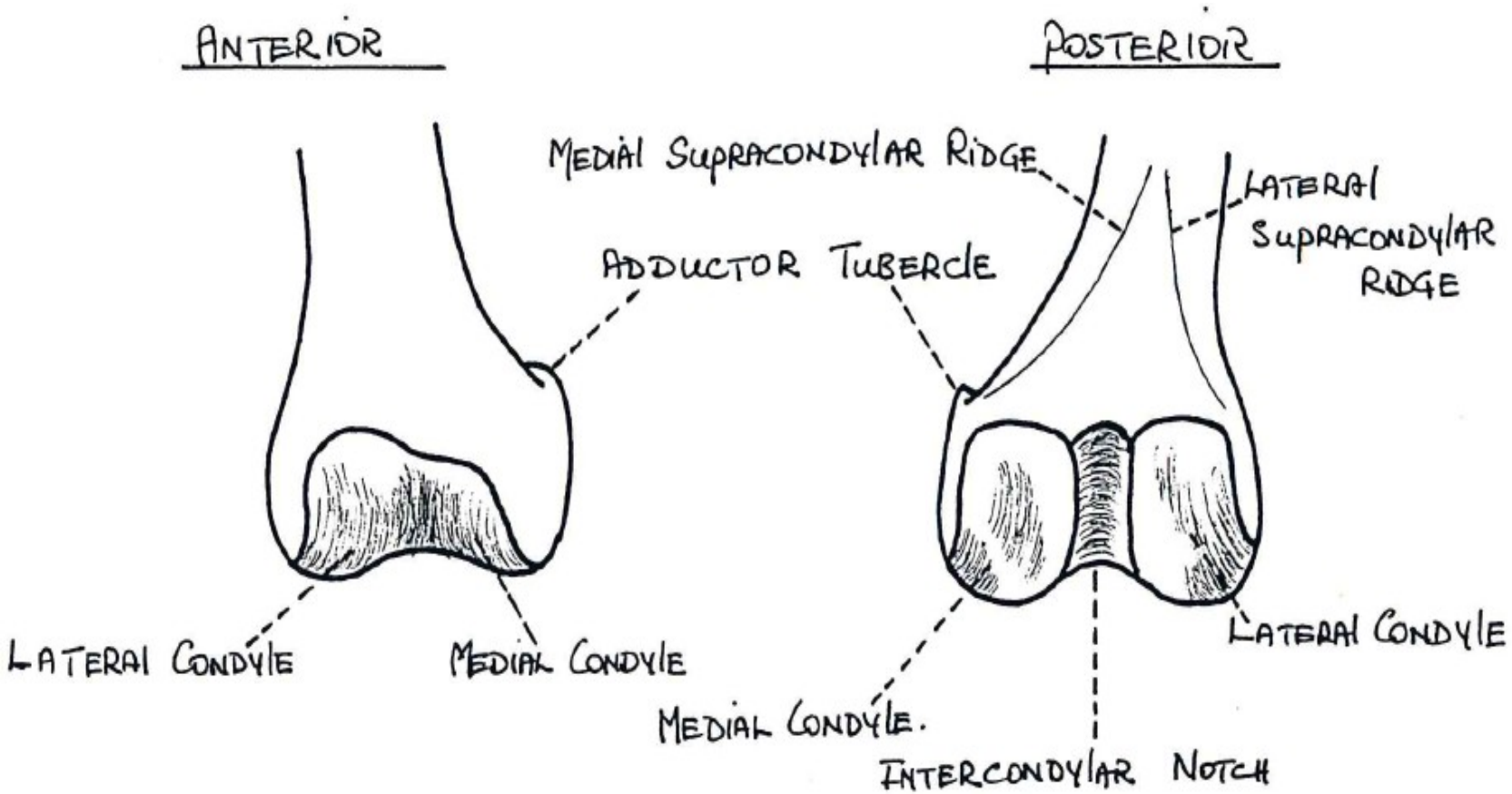
Be familiar with patellar variation (bipartite, alta and baja)

Apply anatomy to a “Baker’s cyst“

Align focal knee pain with anatomical structures

Describe how knee arthroscopy is performed

Understand how traumatic injury to the knee can lead to an “unhappy triad”

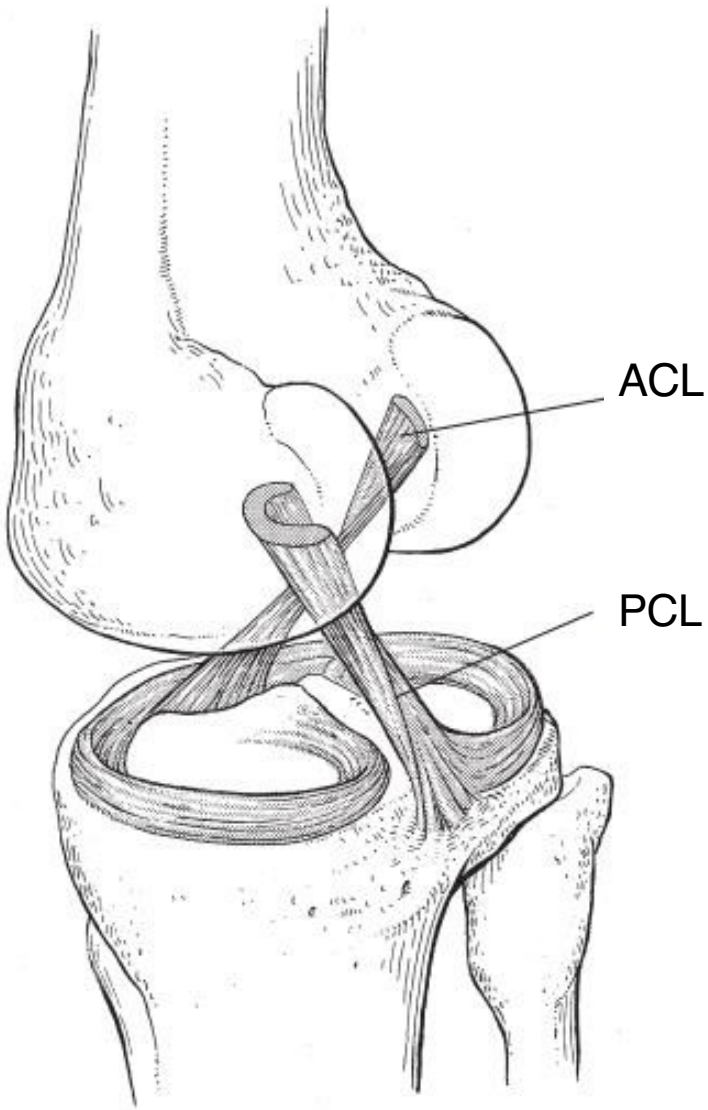

ANTERIOR
POSTERIOR
MEDIAL SUPRACONDYLAR RIDGE
LATERAL SUPRACONDYLAR RIDGE
ADDUCTOR TUBERCLE
LATERAL CONDYLE
MEDIAL CONDYLE
MEDIAL CONDYLE.
LATERAL CONDYLE
INTERCONDYLAR NOTCH

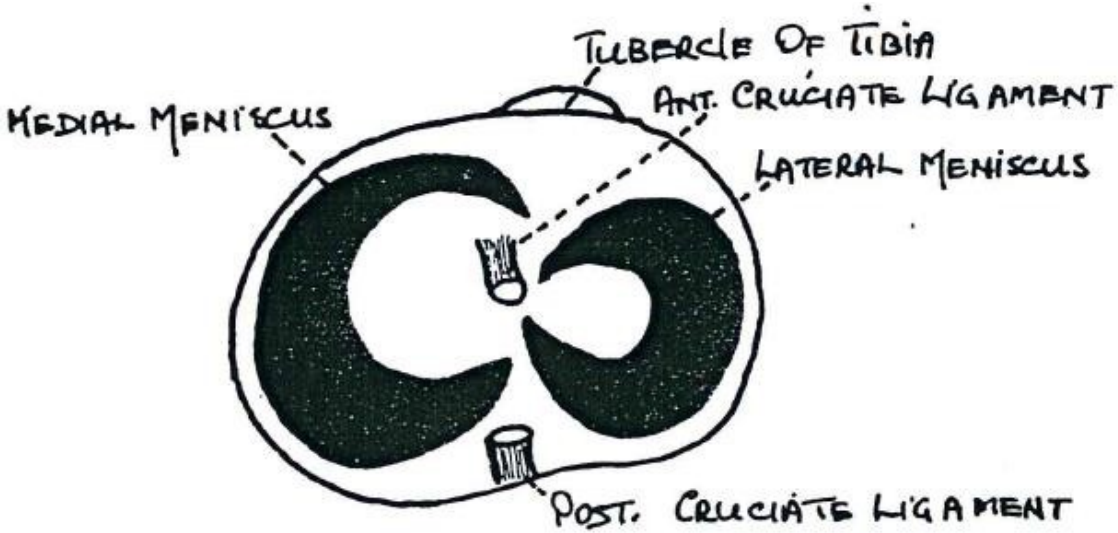
The triceps surae*

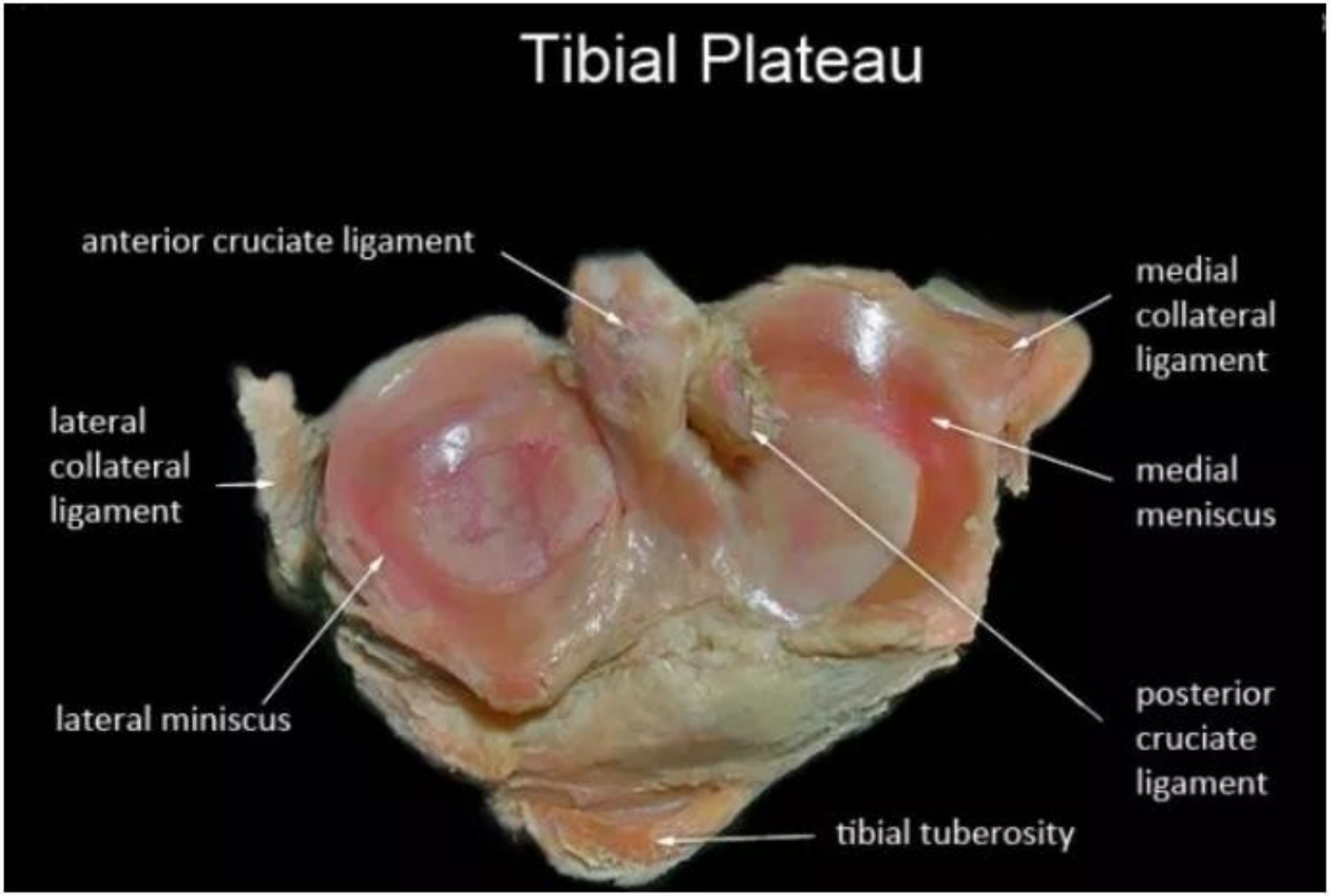
Medial gastrocnemius
Lateral gastrocnemius
Soleus



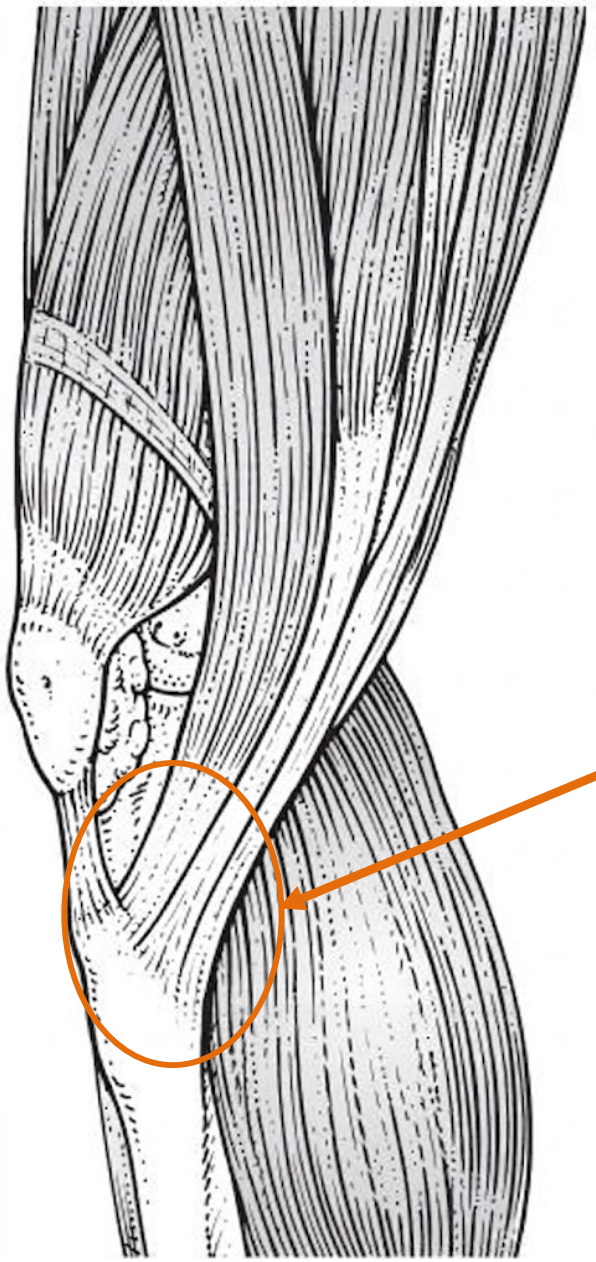
*Sural: relating to the calf of the leg


ACL
PCL

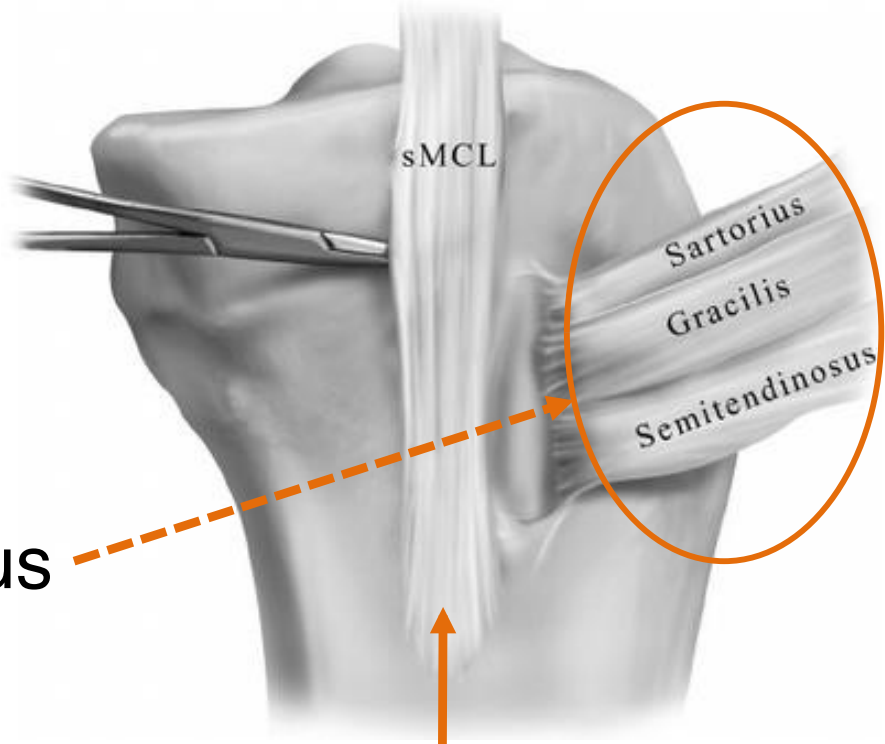

TUBERCLE OF TIBIA
ANT. CRUCIATE LIGAMENT
MEDIAL MENISCUS
LATERAL MENISCUS
POST. CRUCIATE LIGAMENT


Tibial Plateau
anterior cruciate ligament
medial collateral ligament
lateral collateral ligament
medial meniscus
lateral miniscus
posterior cruciate ligament
tibial tuberosity

Medial view of Right Knee

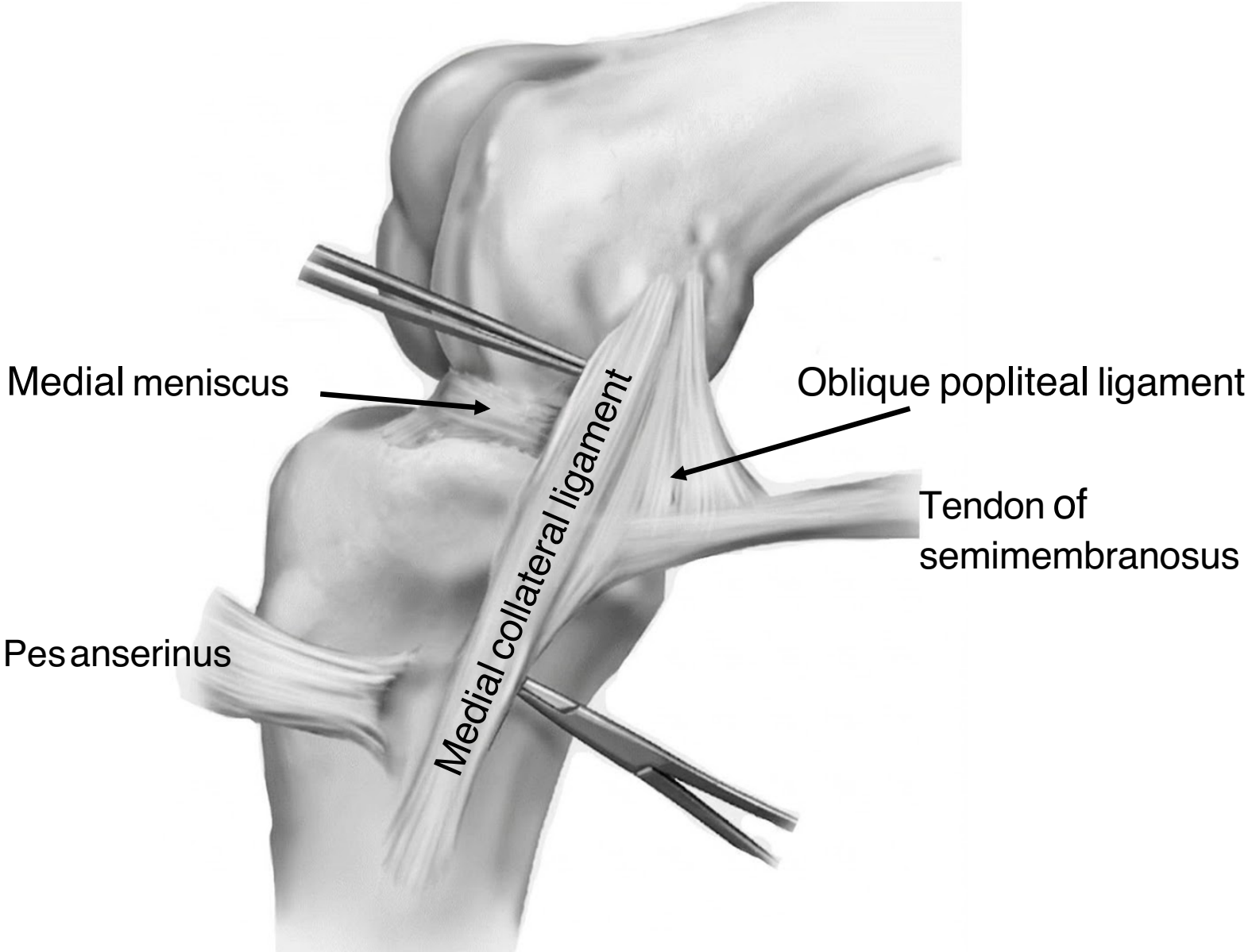


Pes anserinus

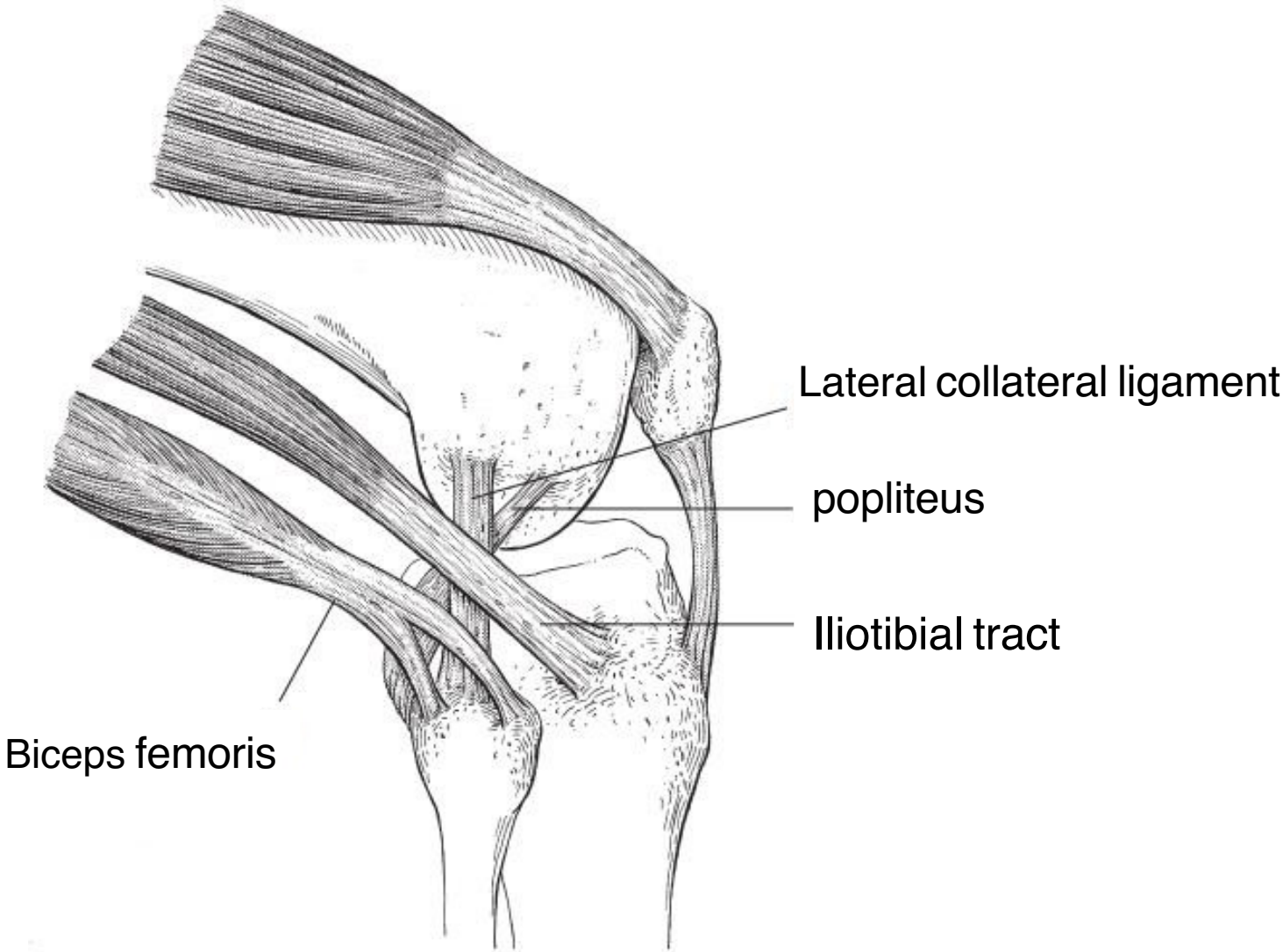


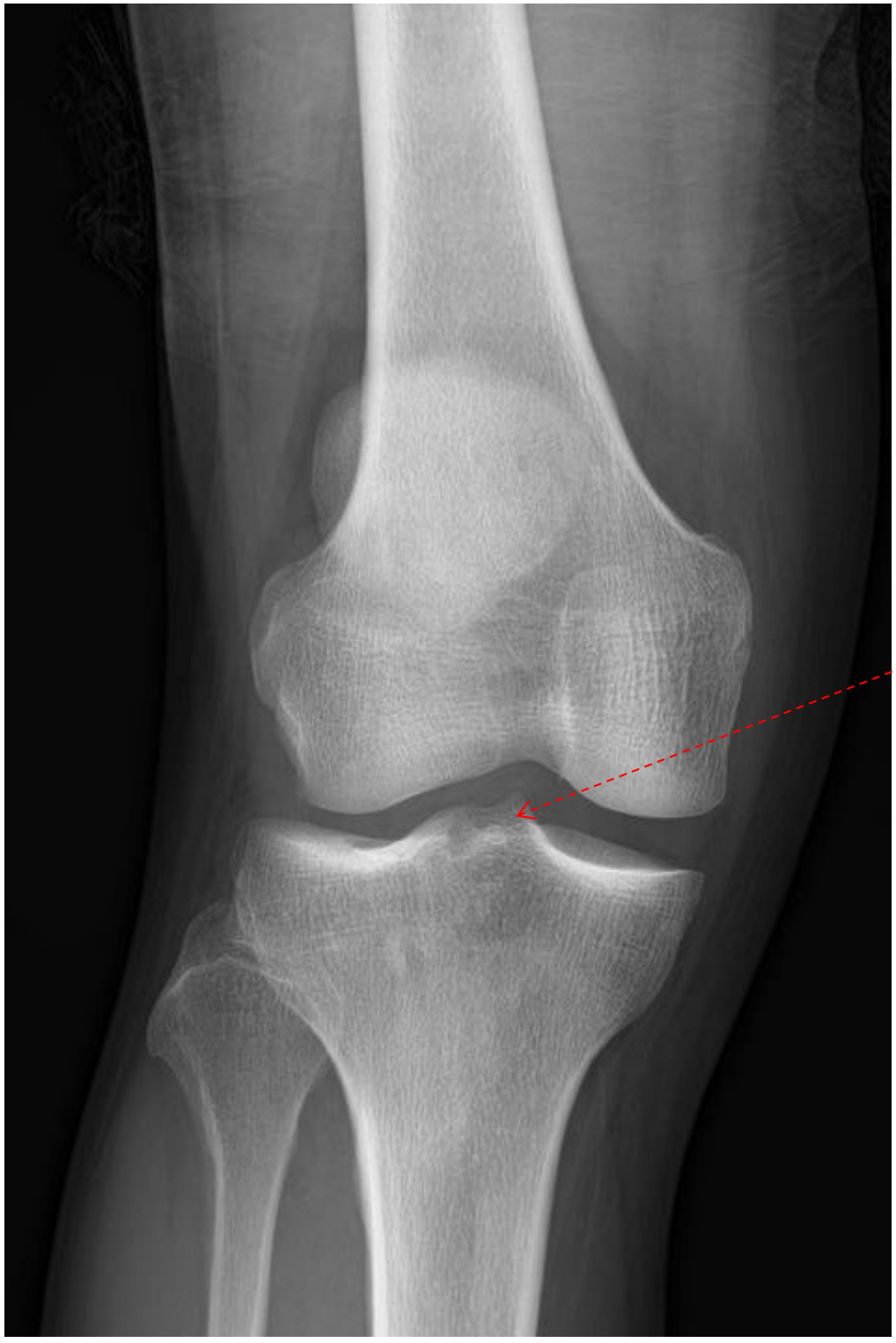
Medial collateral ligament

Medial view of Right Knee

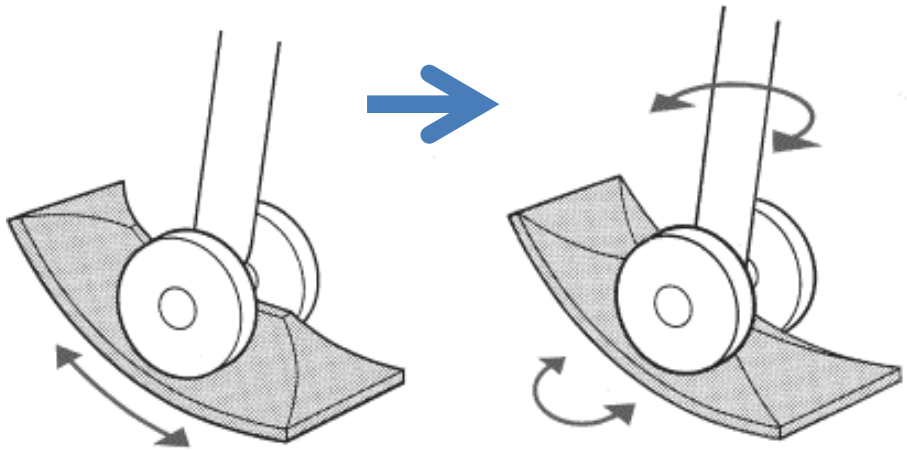


Lateral view of Right Knee



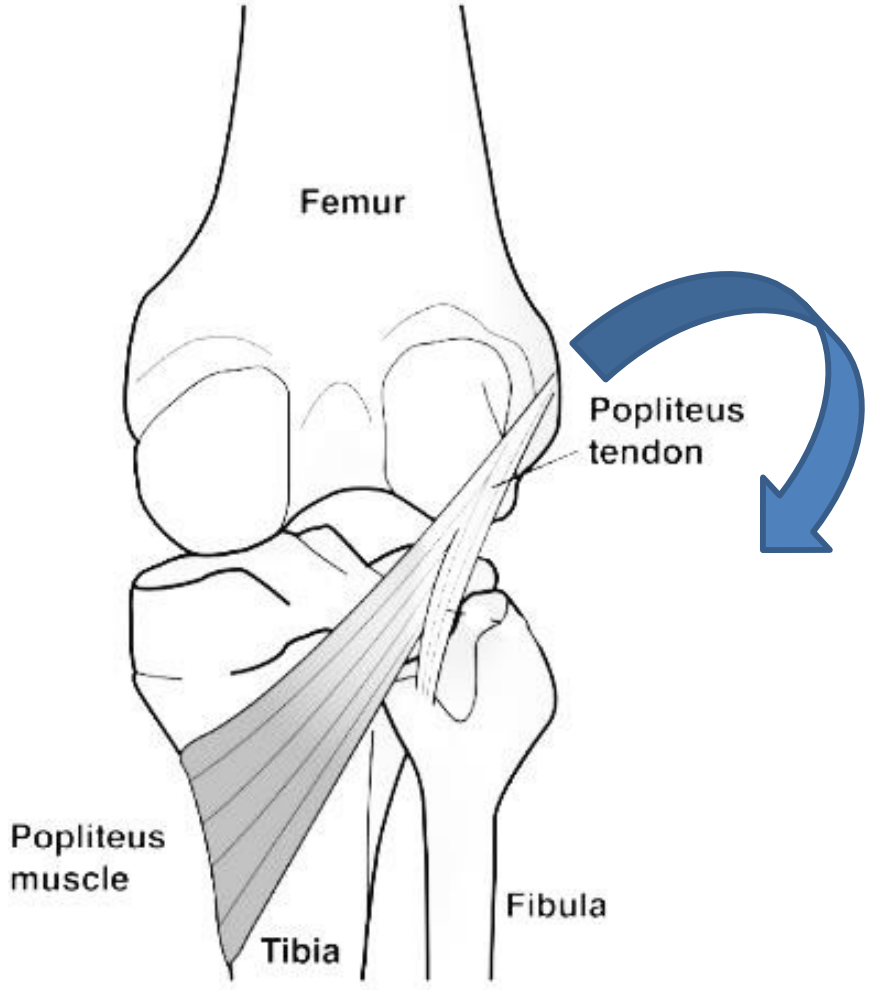

Intercondylar eminence

Flattening the intercondylar rail turns a simple hinge joint into one with rotation

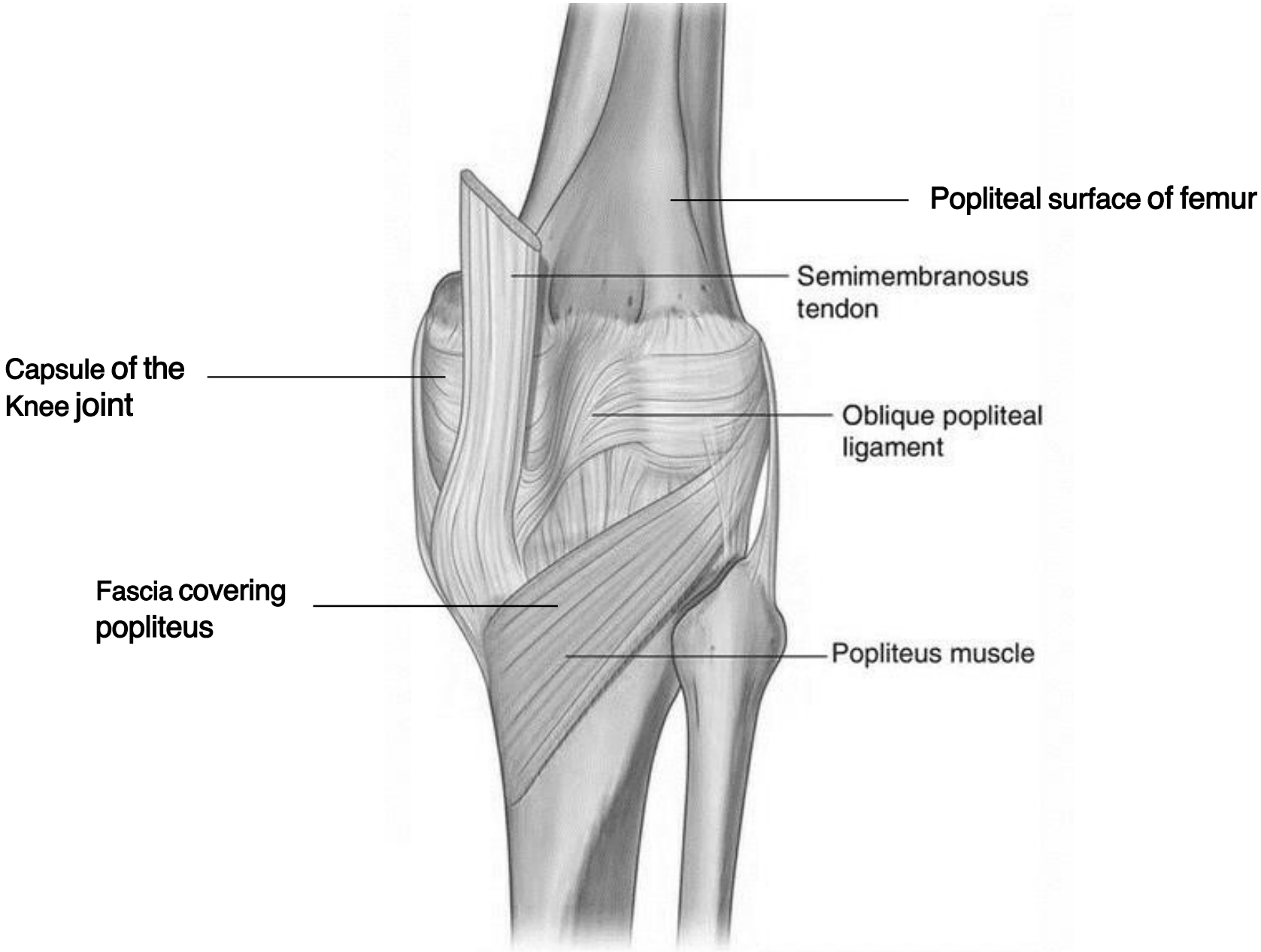


The reduced tibial intercondylar eminence becomes a pivot point

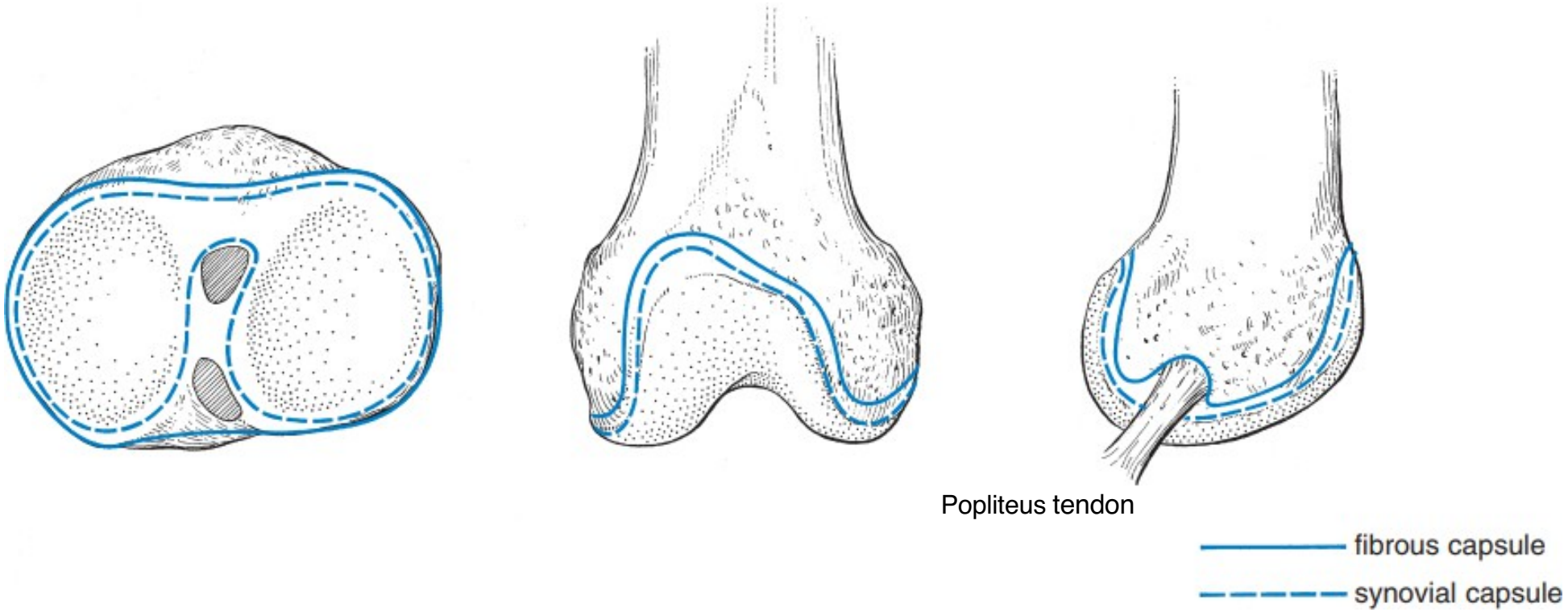
Action of popliteus



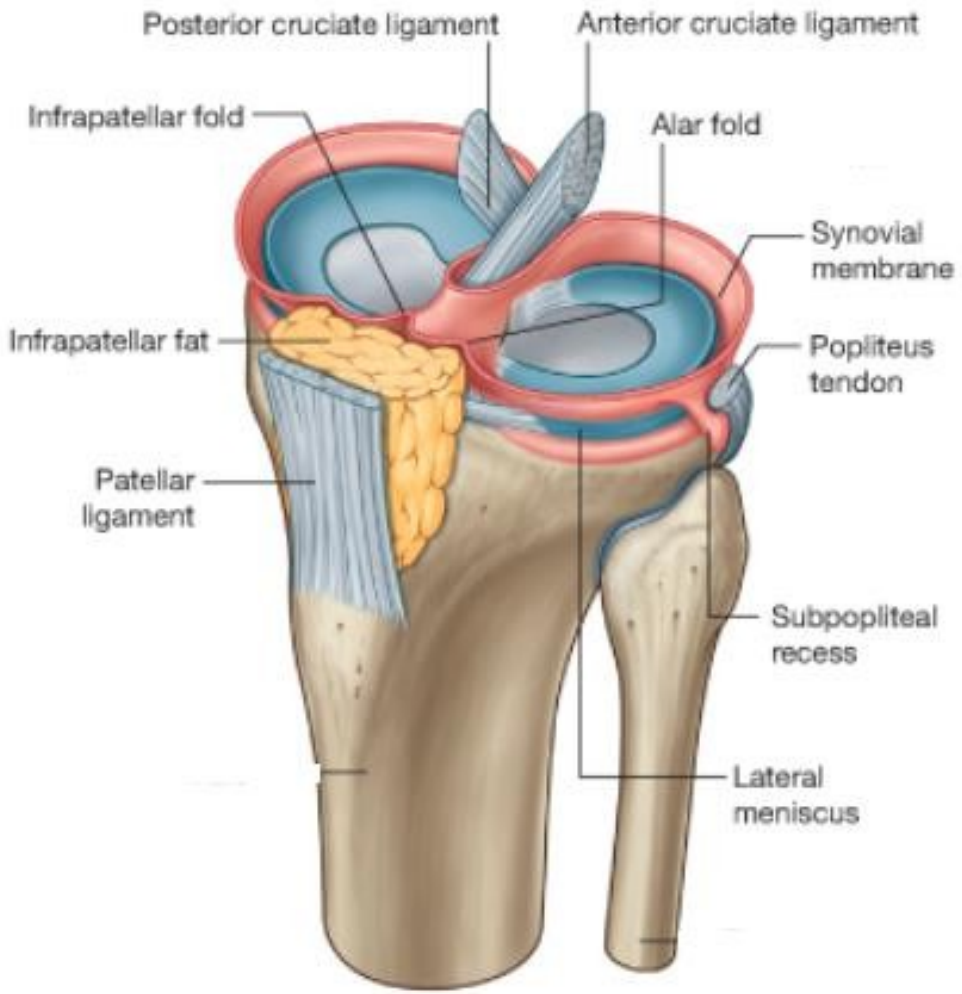
Floor of the popliteal fossa



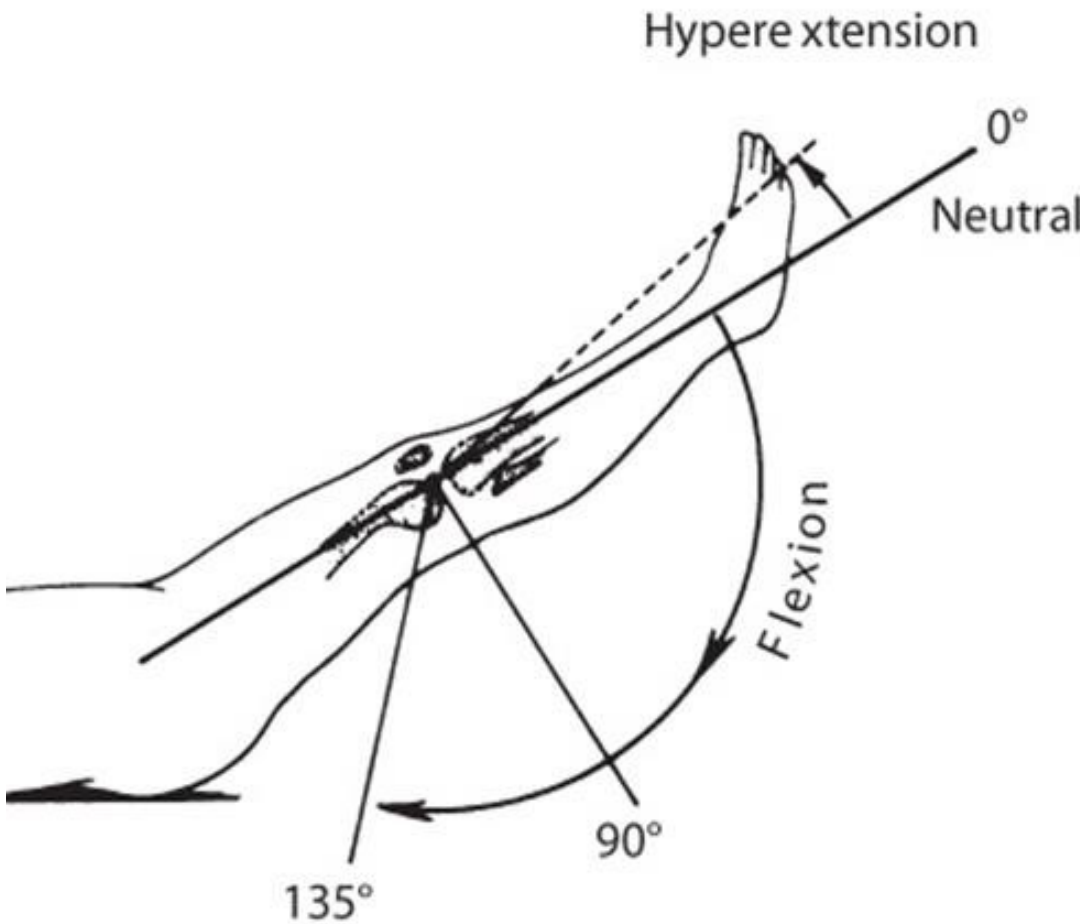
The cruciate ligaments are not bathed by synovial fluid



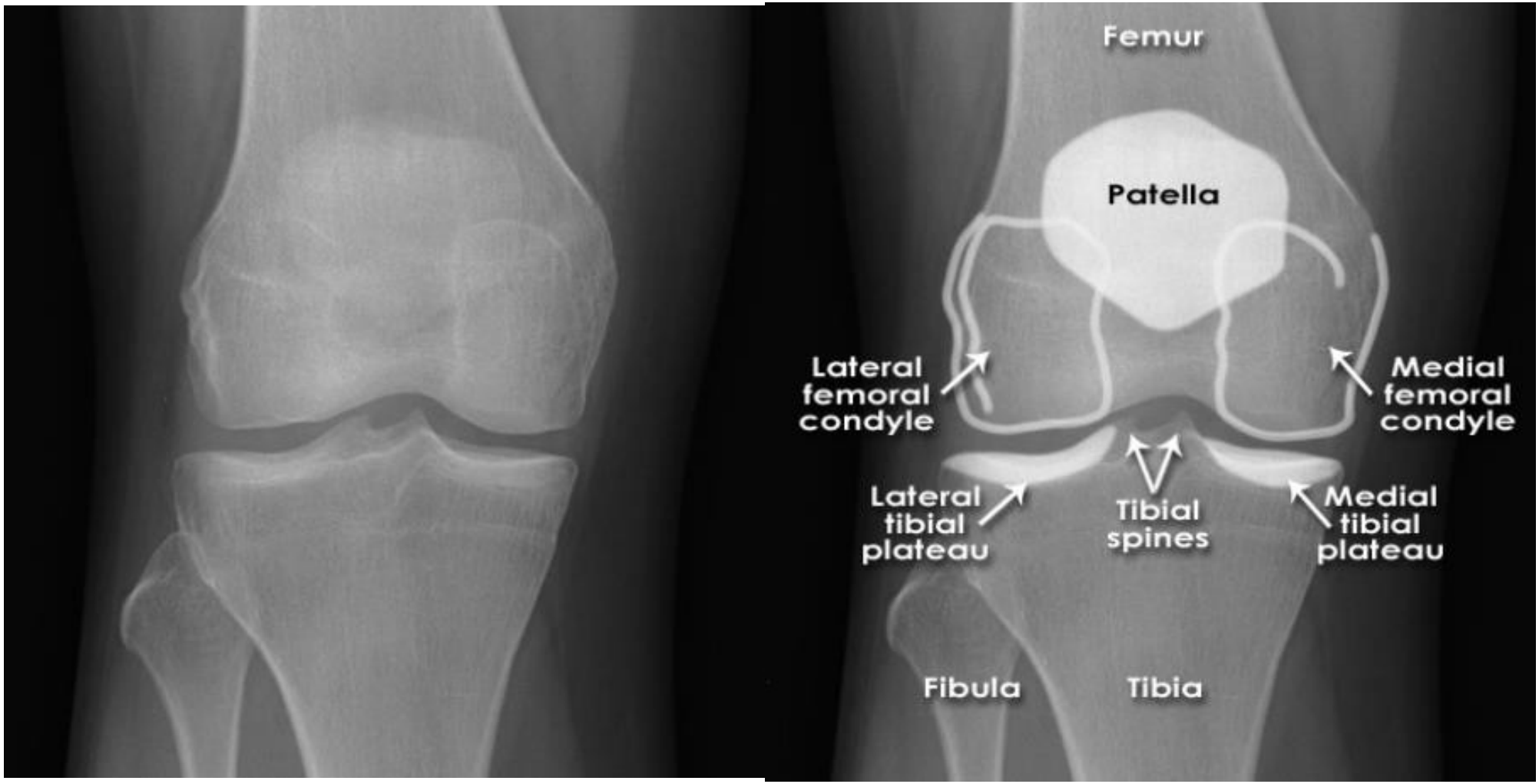
Red synovial membrane



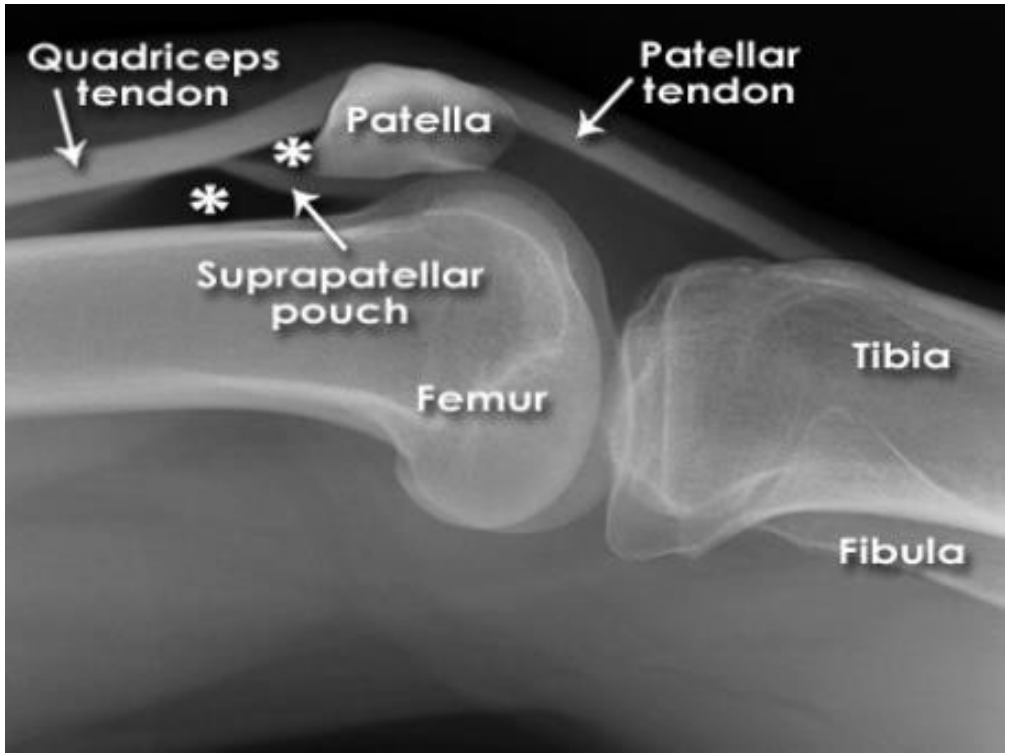
Gray's anatomy for students


Hypere xtension
0°
Neutral
Flexion
90°
135°

Radiology of the Knee

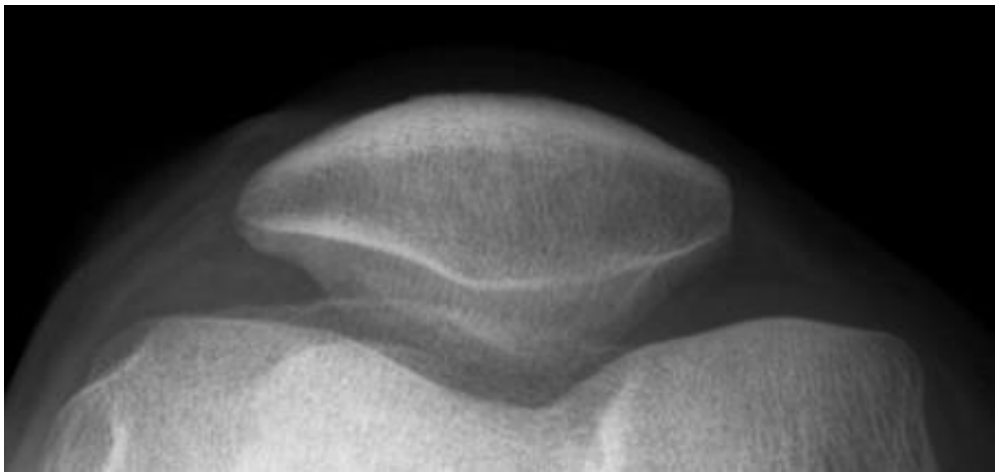

Femur
Patella
Lateral femoral condyle
Medial femoral condyle
Lateral tibial plateau
Tibial spines
Medial tibial plateau
Fibula
Tibia

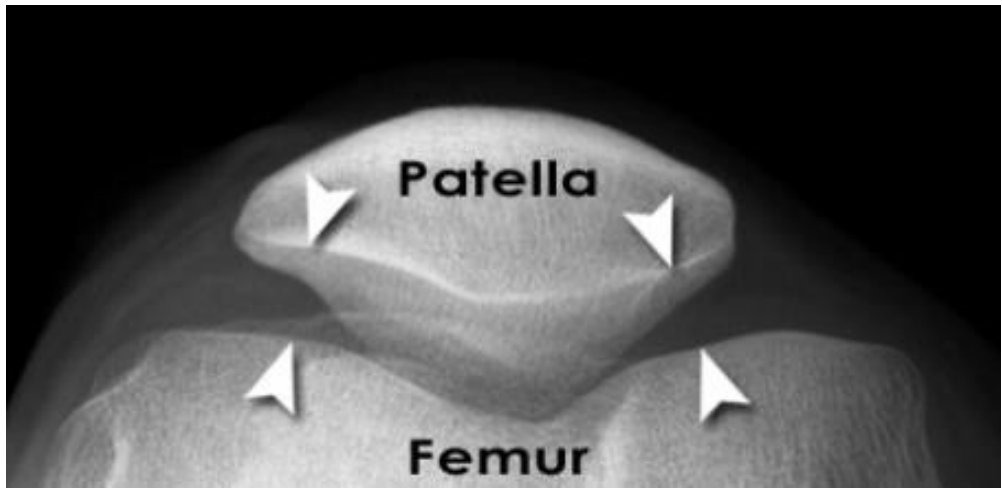



Quadriceps tendon
Patellar tendon
Patella
Suprapatellar pouch
Tibia
Femur
Fibula

* = fat pads

Skyline view of patella (If patient can flex knee)

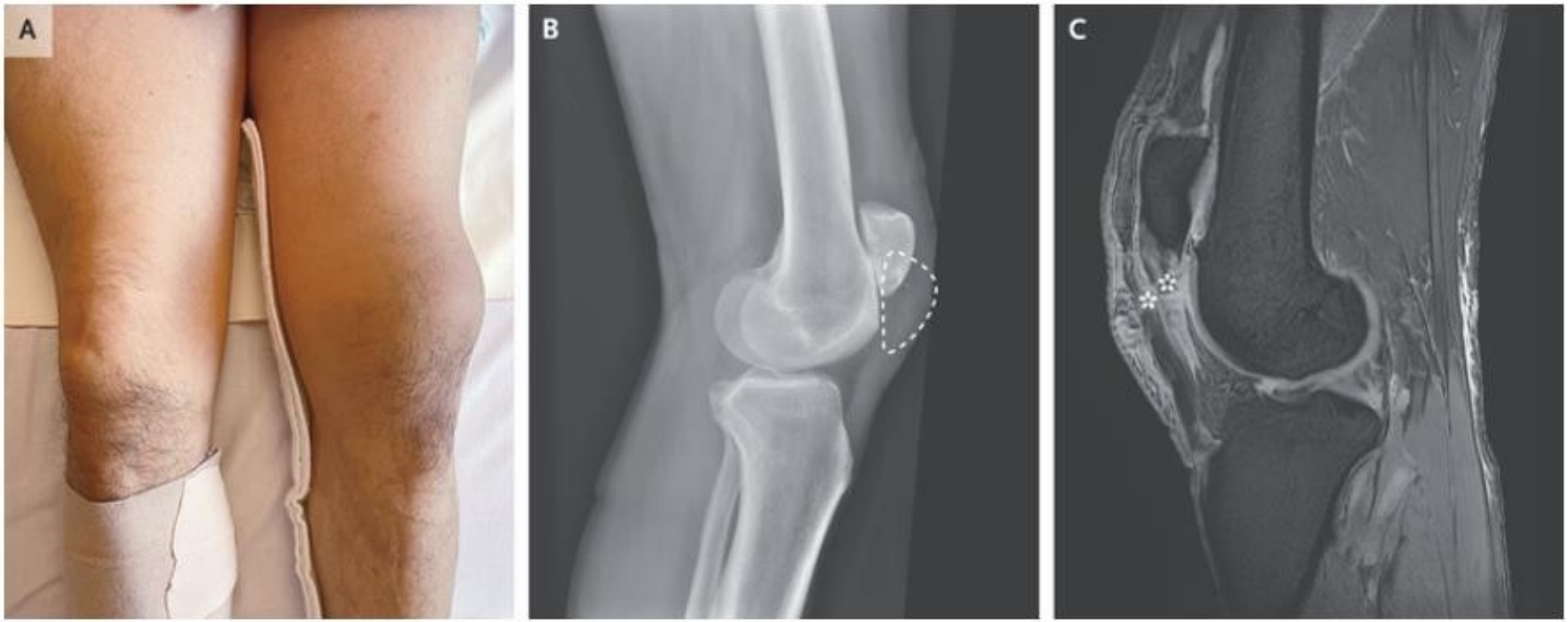




Arrows= normal patello- femoral compartment spacing

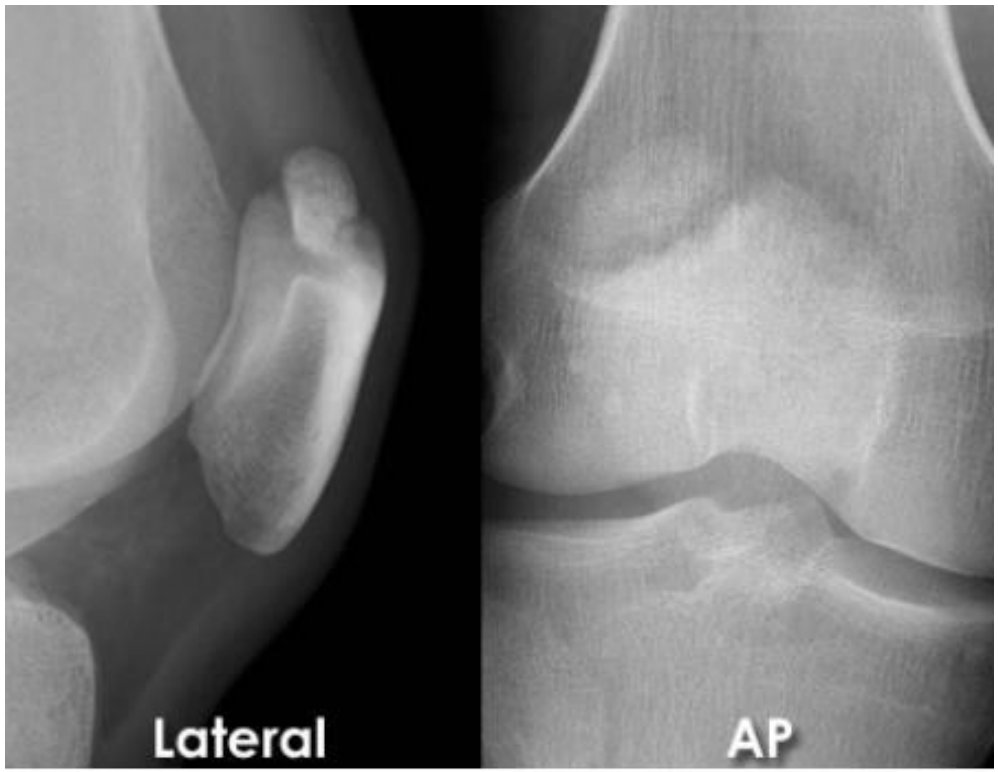
Patella Alta

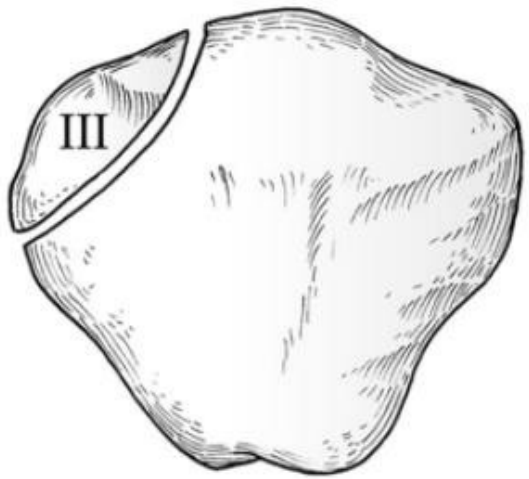
Yoshitaka Ishiguro, M.D., and Akira Kuriyama, M.D., M.P.H., Ph.D.



The NEW ENGLAND JOURNAL of MEDICINE

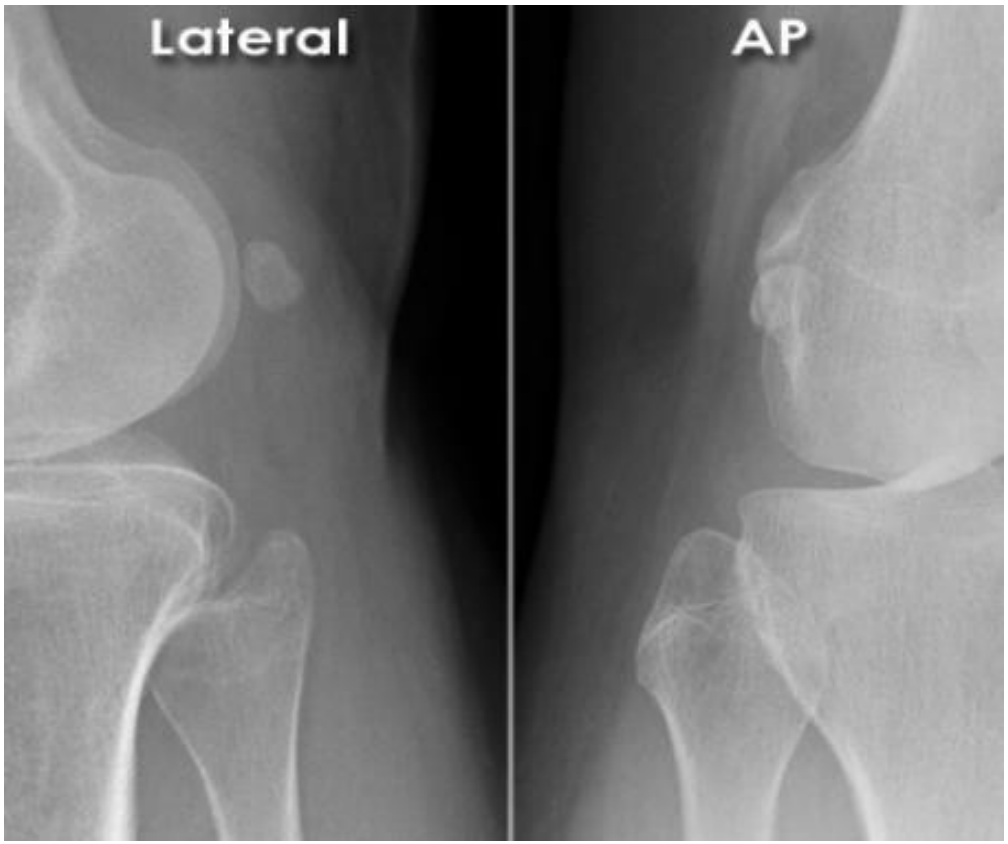
Normal bi partite patella





Superolateral pole
Is common site
Males mainly
50% are bilateral

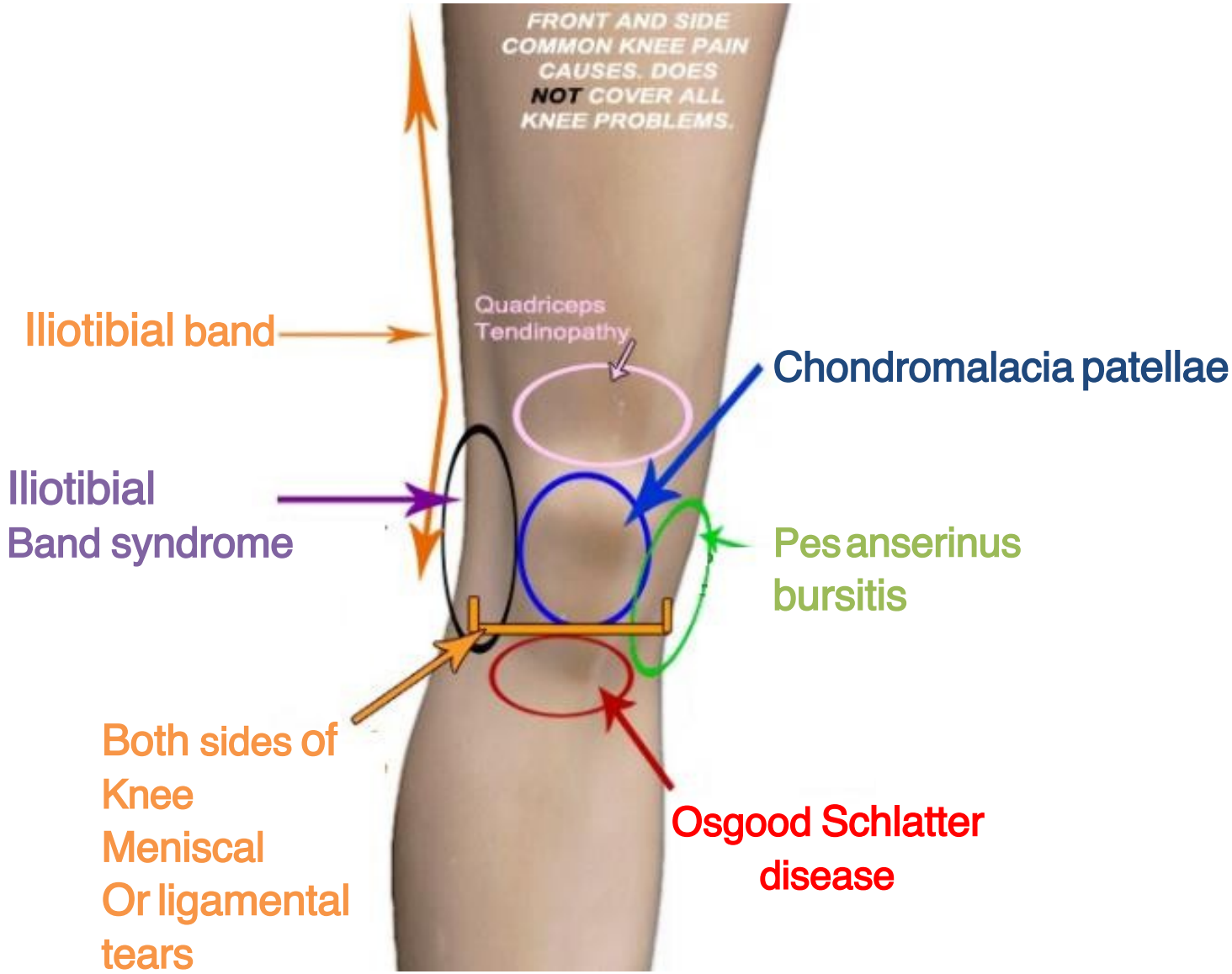
Normal knee with fabella



Fabella is a sesamoid bone in the lateral head of the gastrocnemius muscle

Clinical anatomy of the knee

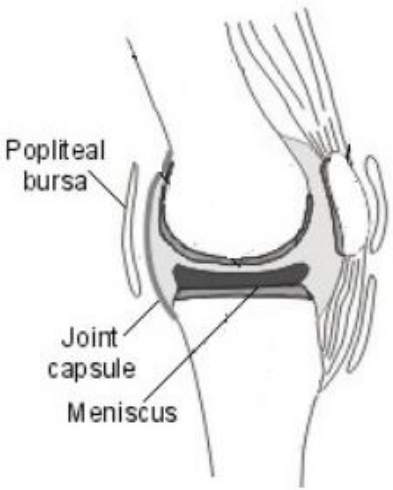
Anterior and side Knee Pain



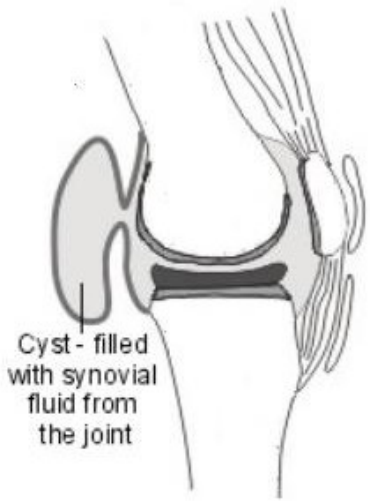
Baker’s cyst



Two types of Baker's cysts

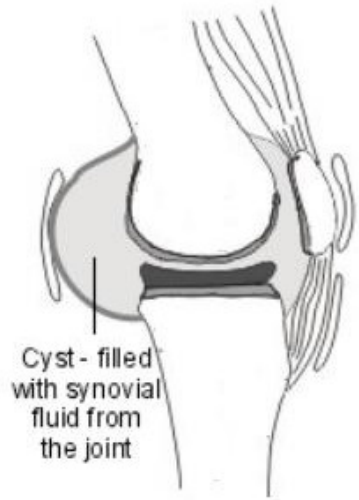


Normal



Primary cyst

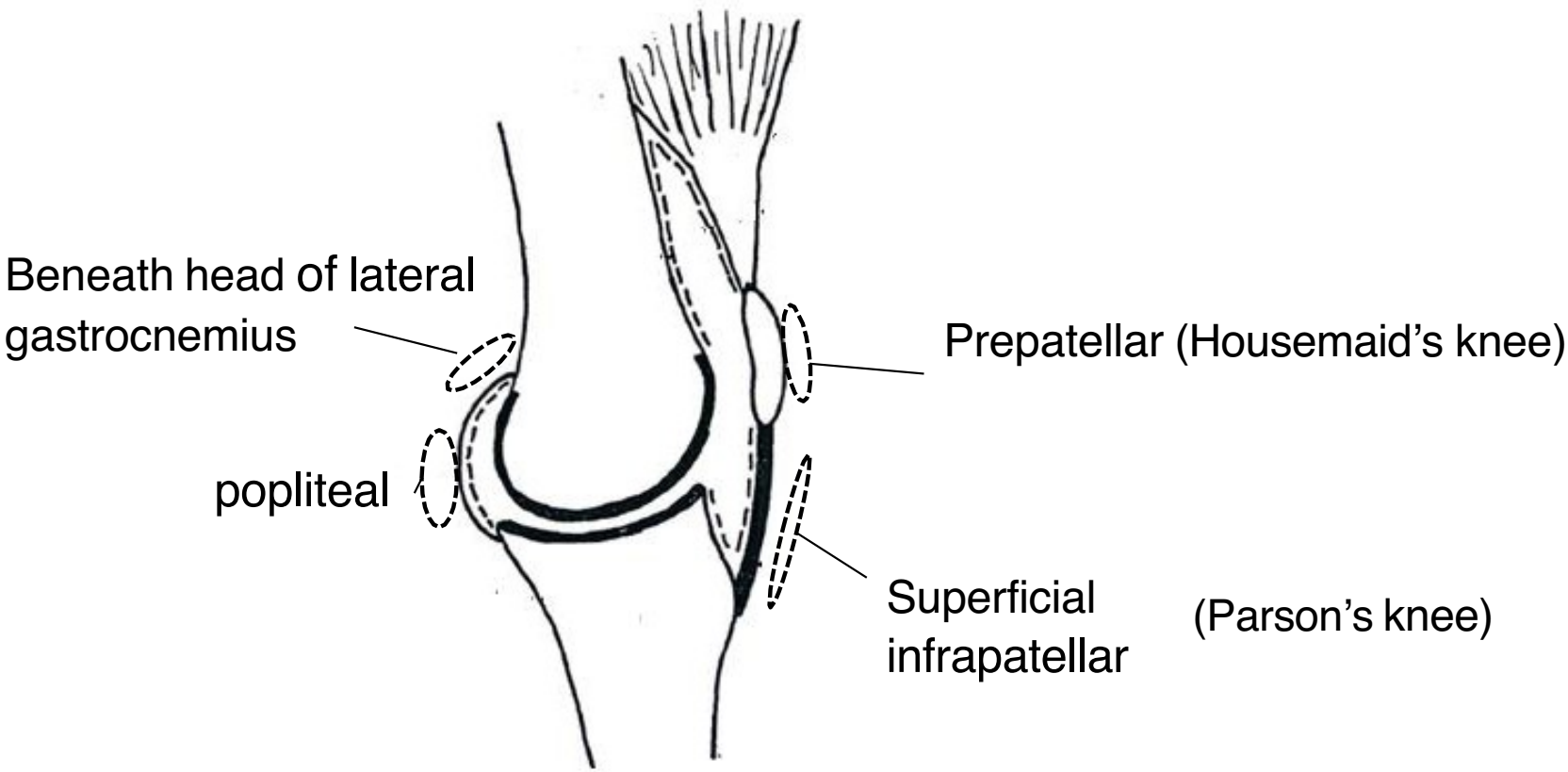
Seen In Children
May not communicate
With knee joint



Secondary cyst

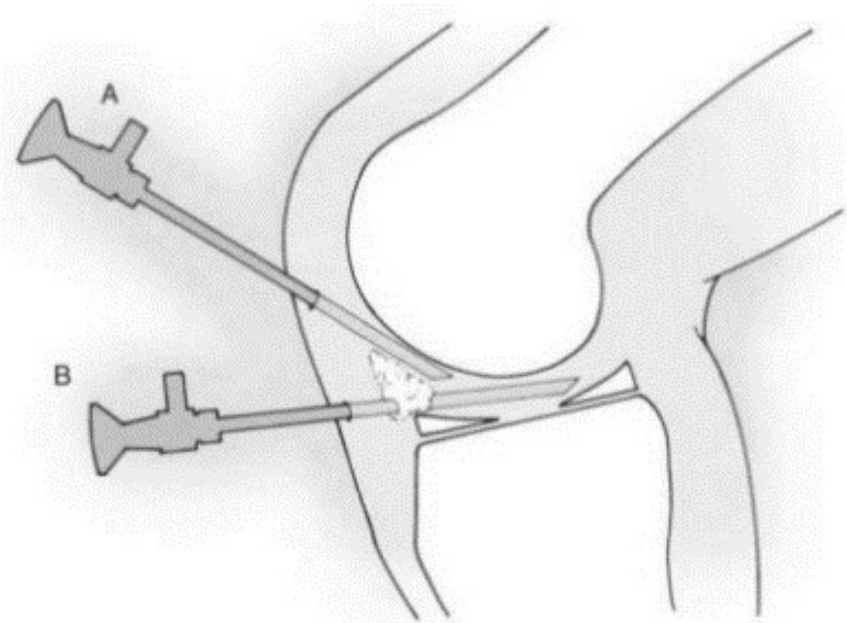
Most Common
Communicates
With knee joint

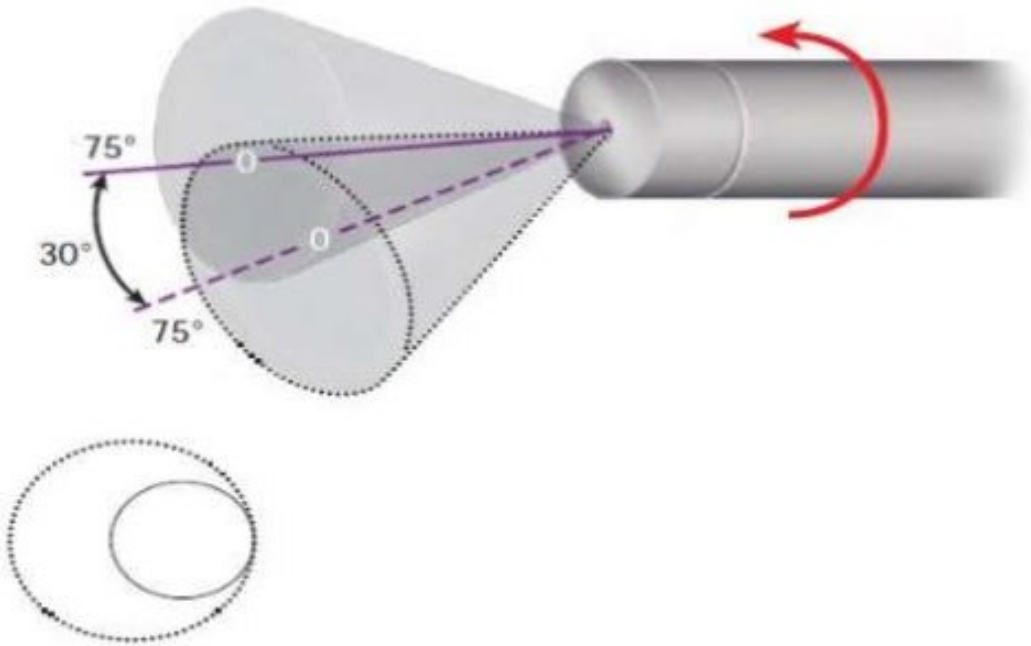
Bursae of the knee



Arthroscopy of the knee







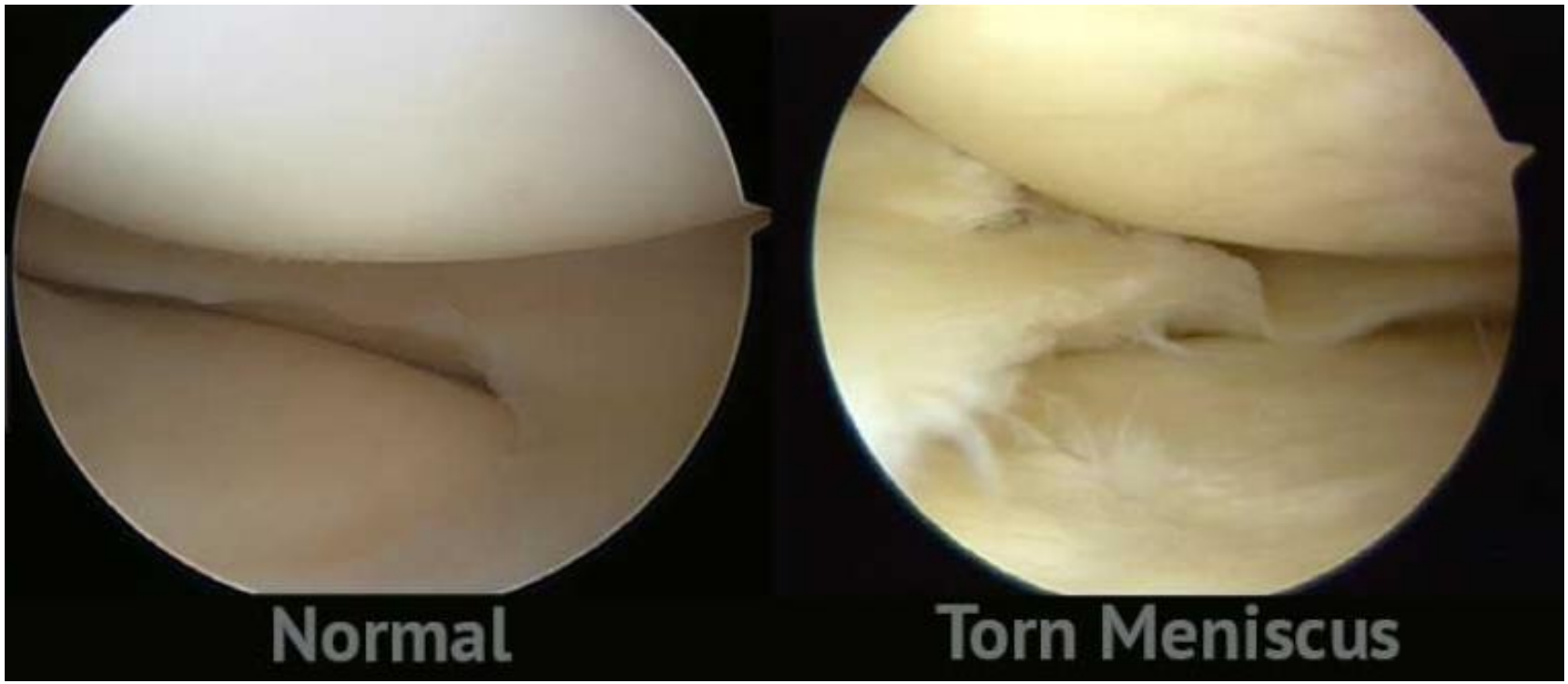
Rotation of arthroscope with 30-degree angle of inclination, which causes scanning effect that increases field of view by about three times.

IRRIGATION SYSTEMS



Two bags of Lactated Ringer's solution

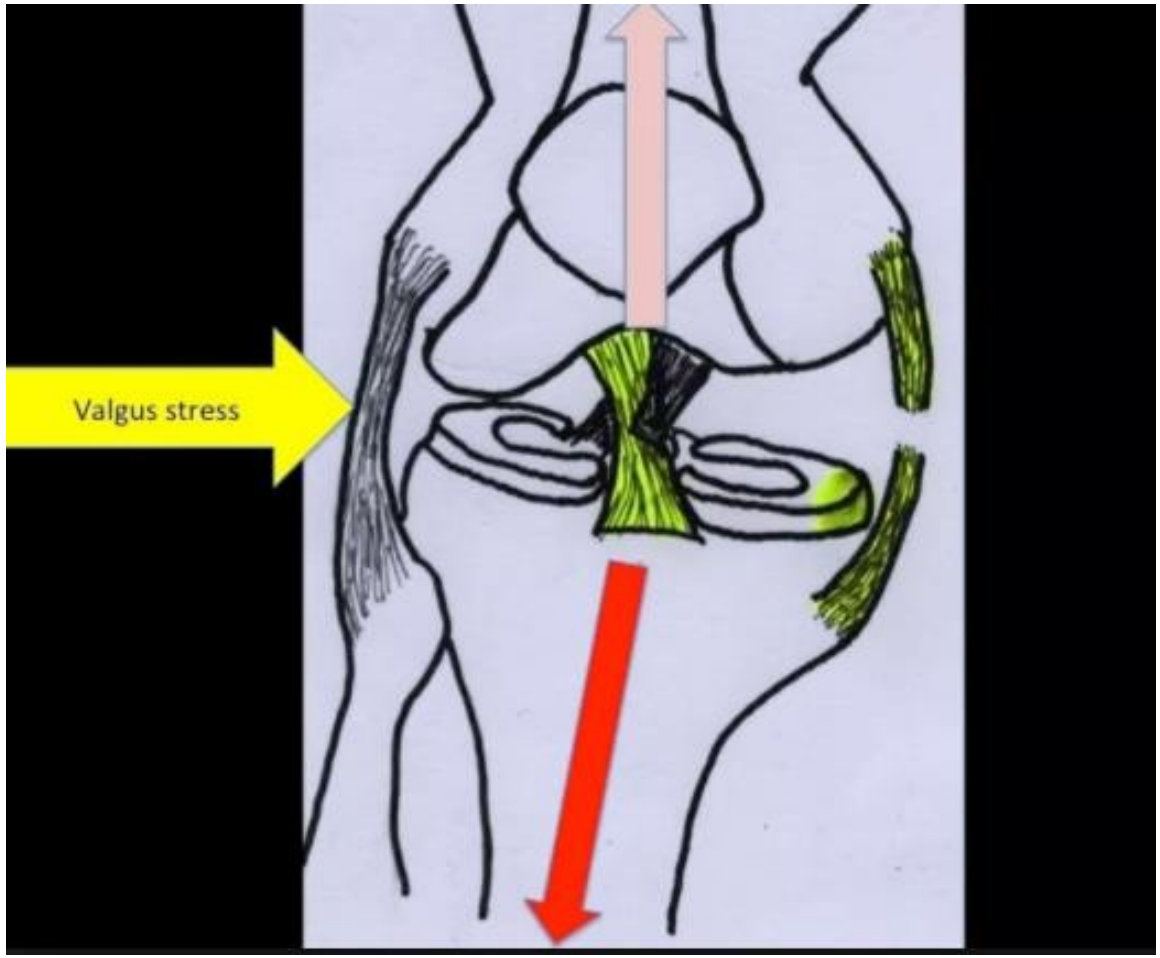
3- 4 feet above joint
Creates 70mmHg approx.


Normal
Torn Meniscus

Traumatic knee injury and the “Unhappy triad”



Unhappy Triad



Tear of :

- Medical meniscus
- Medial collateral ligament
- Ant. Cruciate ligament

Image Challenge

November 02, 2023



38 year old man with end stage kidney disease has difficulty extending his lower limbs And walks with flexed knees